Springfield M1A / M14 Rifle

Springfield Armory / US Military • U.S. Rifle 7.62mm M14 & Springfield M1A National Match (.308 Win)

Manufacturer:	Springfield Armory / US Military	**Category:**	Rifles & Carbines
Action Type:	Short-Stroke Gas Piston Semi-Automatic	**Era / Service:**	Cold War Battle Rifle to Modern DMR (1959–Present)
Calibers:	7.62x51mm NATO, .308 Winchester	**Covered Models:**	M1A Standard, M1A Loaded, M1A National Match, M1A Scout Squad (18"), SOCOM 16 (16.25")

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Gas Cylinder Unitizing (Screw-and-Glue / Welding) • Castle Nut Pliers & Set Screw Service • Spindle Valve Alignment (Off vs On) • Stock Ferrule Clearance Check • Op-Rod Guide Staking

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
GAS SYSTEM DESIGN	Tappet expansion-cutoff short-stroke gas piston (White expansion gas system)
UNITIZING GAS CYLINDER	Front band and gas cylinder locked together permanently (screw-and-glue or TIG weld) so front band does not shift on barrel under heat
STOCK FERRULE CLEARANCE	Front handguard ferrule must rest on lower stock ferrule lip with downward tension (approx 3–5 lbs draw tension)
SPINDLE VALVE	Rotatable gas valve for rifle grenades; slot MUST align parallel with barrel for semi-auto cycling

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Rifle Becomes Single Shot (Bolt Stays Closed After Firing) **Rifle fires with full power, but bolt remains locked forward; no gas reaches operating piston**	• Spindle valve on gas cylinder accidentally rotated 90° into grenade-launching closed position.	**Immediate Action: Look at right side of gas cylinder. Use screwdriver or rim of cartridge to turn spindle valve slot so it is PARALLEL with the bore line, restoring gas flow to piston.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Springfield M1A / M14 Armorer Maintenance & Accuracy Secrets	Brownells Gunsmithing Bench	Complete walk-through of the M1A: gas piston cleaning without removing cylinder, gas cylinder unitizing, bolt roller greasing, and stock draw pressure.